(ii) from acetylene.

Acetylene on passing through a red –hot tube trimerises to give benzene. We have already studied this concept in polymerization of alkynes.

$$3\ CH{\equiv}CH \xrightarrow[873\ K]{\text{Red hot iron tube}} C_6H_6$$

Acetylene — Benzene

(iii) Laboratory Methods Of Preparing Benzene And Toluene

(a) Decarboxylaation Of Aromatic Acid.

When sodium benzoate in heated with sodalime, benzene vapours distil over.

$$C_6H_5COONa + NaOH \xrightarrow{CaO} C_6H_6 + Na_2CO_3$$

sodium benzoate — benzene

(b) Preparation Of Benzene From Phenol

When phenol vapours are passed over zinc dust, then it is reduced to benzene.

$$C_6H_5OH + Zn \longrightarrow C_6H_6 + ZnO$$

phenol

(c) Wurtz – Fittig Reaction:

When a solution of bromo benzene and iodo methane in dry ether is treated with metallic sodium, toluene is formed.

$$C_6H_5Br + 2Na + ICH_3 \xrightarrow{\text{ether}} C_6H_5{-}CH_3 + NaBr + NaI$$

Bromo benzene — Iodo methane — Toluene

(d) Friedel Craft's Reaction:

When benzene is treated with methyl chloride in the presence of anhydrous aluminium chloride, toluene is formed.

$$C_6H_6 + CH_3Cl \xrightarrow{\text{anhydrous } AlCl_3} C_6H_5CH_3 + HCl$$

Benzene — chloromethane — toluene

Evaluate Yourself

24) How benzene can be prepared by Grignard Reagent

Unit 13 - jagan.indd 210 22-01-2021 12:51:40